

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7
Months\n\nAuthors: Thomas Kwa,
Ben West, Joel Becker, et al. (METR)\n\nSource:
<https://arxiv.org/abs/2503.14499>

A New Metric: 50% Task-Completion Time Horizon

Methodology involves three steps:

1. Assemble a diverse suite of 170 software/research tasks
2. Time skilled humans and measure AI agent success rates
3. Fit logistic model to estimate the task duration at which each AI agent achieves 50% success

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

(97 diverse software tasks), RE-Bench (7 difficult ML research tasks), SWAA (66 short single-step tasks)\nOver 800 human baselines

Task Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of a backtesting tool

AI Task Horizon Grew from 2 Seconds to ~1.8 Hours in 6 Years

Exponential Trend (2019-2025):

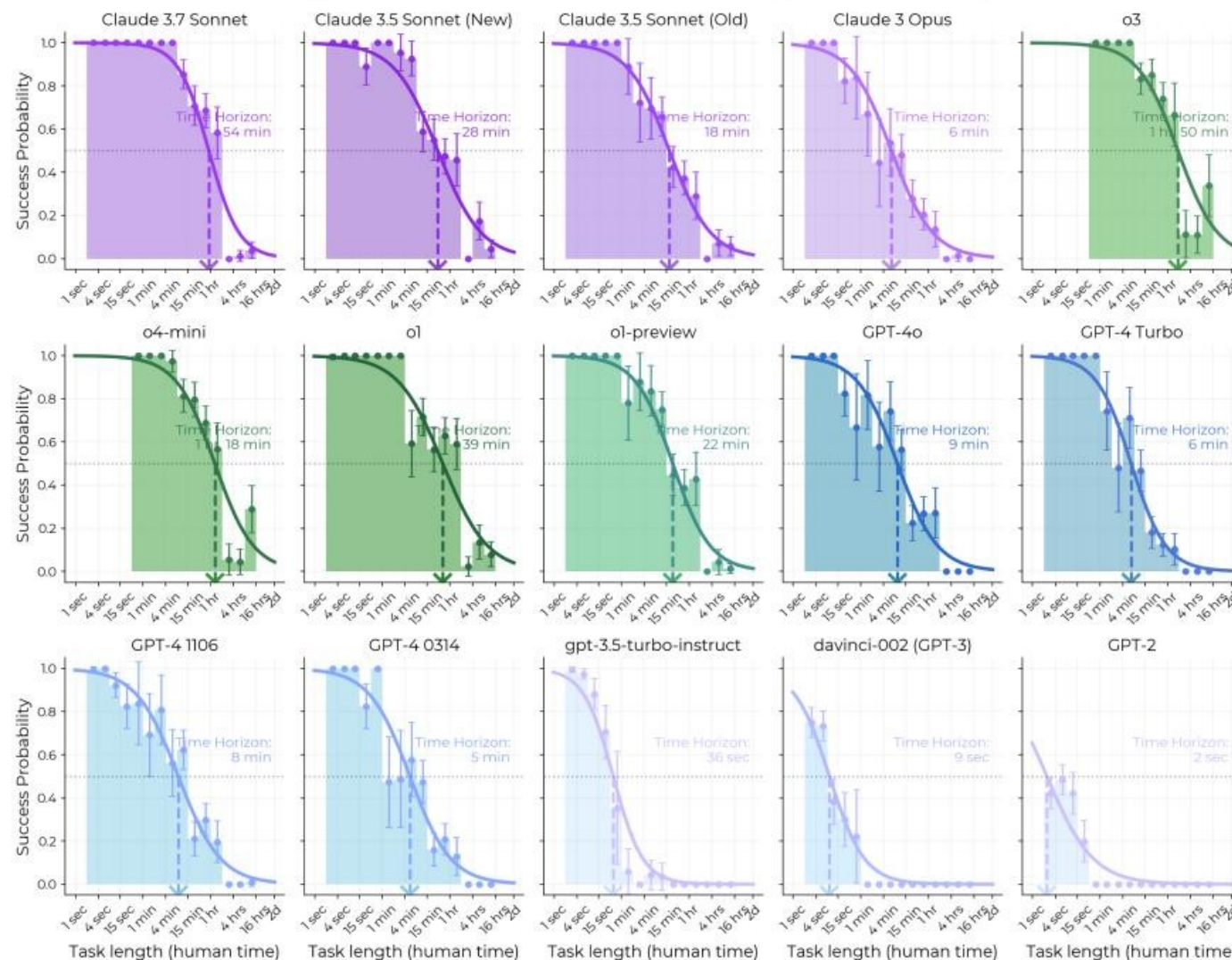
Doubling time: ~207 days (~7 months)

$$R^2 = 0.97$$

GPT-2 (2019): ~2 seconds

GPT-4 0314 (2023): ~5 minutes

o3 (2025): ~1 hour 50 minutes



Success Probability Drops Sharply as Task Length Increases

Per-Model Success Probability:

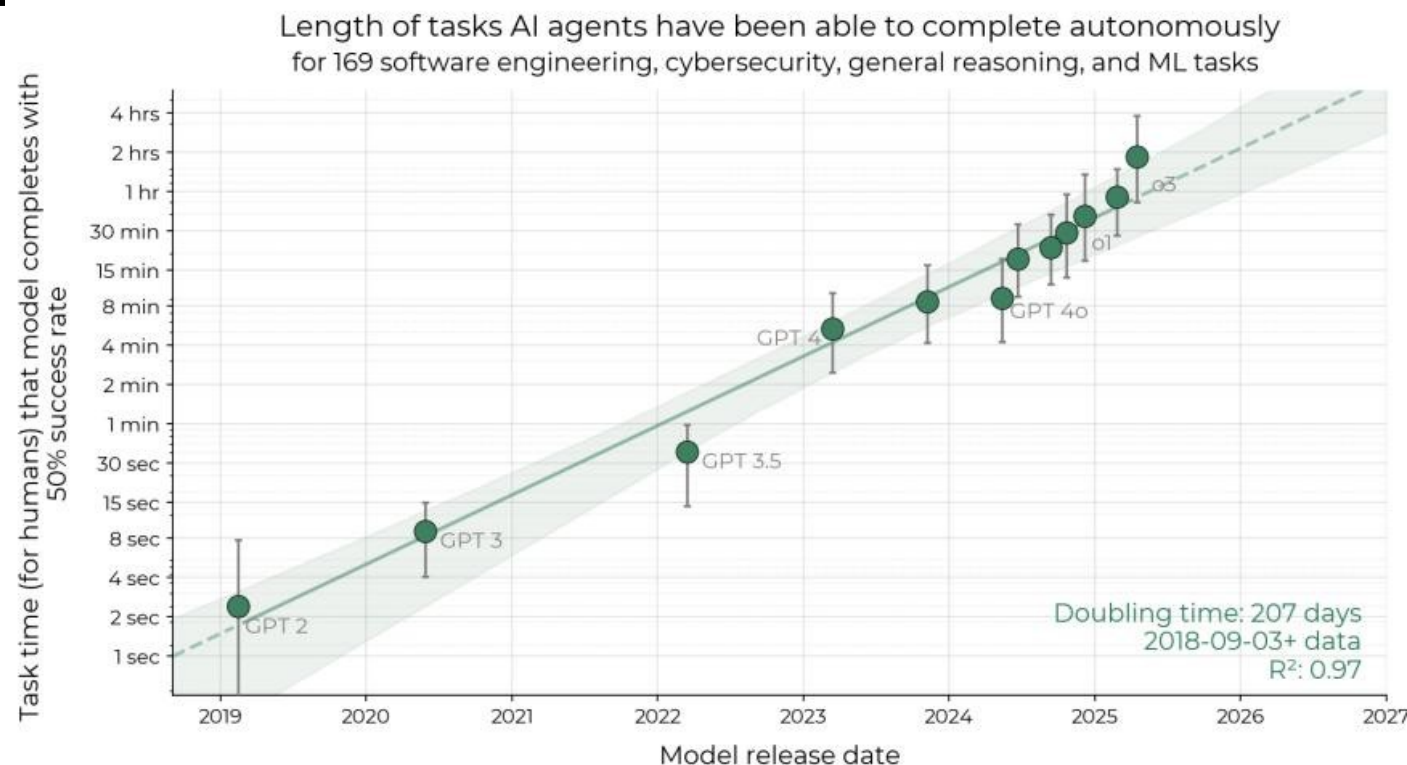
Success rate declines with task length for all models.

Frontier models push the curve rightward:

Claude 3.7 Sonnet: 54 min

o3: 1 hour 50 min

GPT-2: 2 seconds



Reasoning, Tool Use, and Error Recovery Drive Growth

- Key Capability Drivers:
 - Improved logical reasoning
 - Better tool use
 - Greater reliability and self-awareness in task execution
 - Ability to adapt to mistakes
- Limitations:
 - Performance is notably lower on less structured, 'messier' tasks.

Extrapolation Predicts Month-Long Task Automation by 2028-2031

- Forward Extrapolation:
 - If the trend continues, AI reaches >1 month time horizon (167 work hours) between mid-2028 and mid-2031.
 - 2023-2025 horizon growth rate is ~20% faster than the overall 2019-2025 rate, suggesting possible acceleration.
- Caveats:
 - External validity remains an open question.
 - Tasks may not perfectly represent the average

Takeaways: Exponential Autonomous Capability Demands Governance

- 1. 50% time horizon is a robust, intuitive metric for tracking AI autonomy with $R^2=0.97$ over 6 years.
- 2. Doubling every ~7 months with possible acceleration since 2024.
- 3. Current frontier is at ~2 hours of human-equivalent work (o3).
- 4. External validity remains an open question —tasks may not perfectly represent real-world